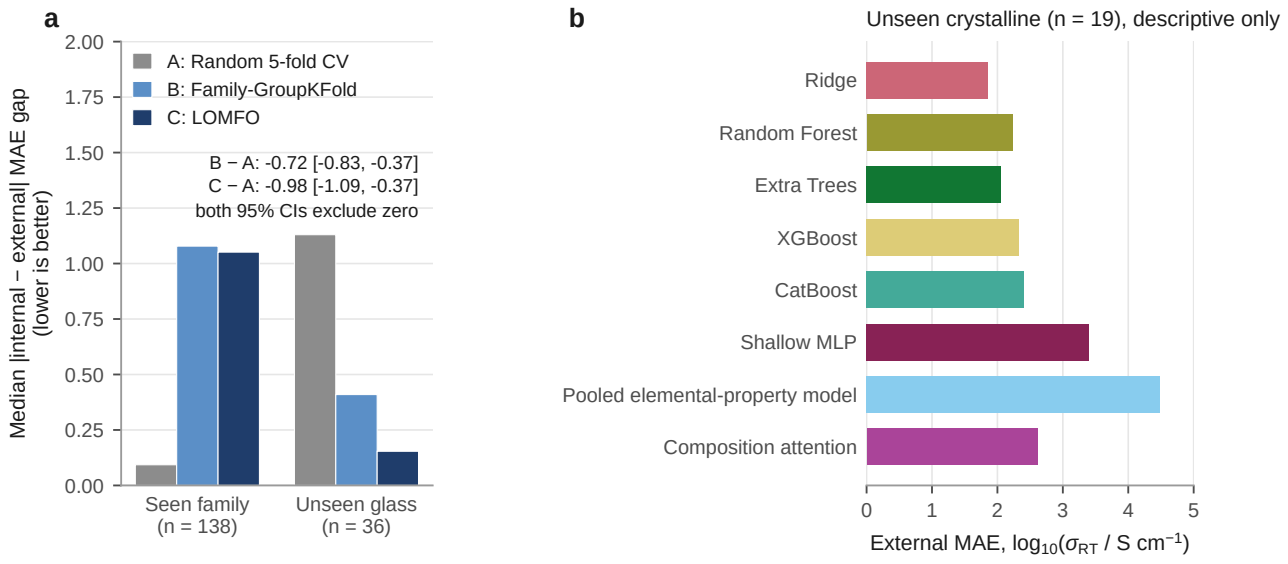
a
A: Random 5-fold CV
B: Family-GroupKFold
C: LOMFO
B − A: -0.72 [-0.83, -0.37]
C − A: -0.98 [-1.09, -0.37]
both 95% CIs exclude zero
Median |internal − external| MAE gap
(lower is better)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Seen family
(n = 138)
Unseen glass
(n = 36)
b
Unseen crystalline (n = 19), descriptive only
Ridge
Random Forest
Extra Trees
XGBoost
CatBoost
Shallow MLP
Pooled elemental-property model
Composition attention
0
1
2
3
4
5
External MAE, log10(σRT / S cm−1)